Journal of Autism and Developmental Disorders
https://doi.org/10.1007/s10803-018-3502-4

ORIGINAL PAPER

Using Peer-Mediated LEGO® Play Intervention to Improve Social Interactions for Chinese Children with Autism in an Inclusive Setting

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Abstract

The purpose of this study was to examine the effects of a peer-mediated LEGO® play intervention on improving social skills for children with ASD in an inclusive preschool in China. Three boys with ASD and 13 typically developing children participated in this study. A multiple-probe across participants design was used. The intervention consisted of LEGO® construction activities incorporated with peer-mediated strategies for one child with ASD and two typically developing peers. The intervention sessions were conducted two sessions per week with a total of 28–31 sessions for each participant. Results indicated that all three children with ASD increased their social initiations and responses following the completion of the intervention. Social validity was also obtained.

Keywords Autism spectrum disorder · LEGO® play · Social skills · Peer mediation · Inclusive education

Introduction

Autism spectrum disorder (ASD) is a neurodevelopmental disorder characterized by social communication impairments and by restricted interests and repetitive behaviors (American Psychiatric Association 2013). One of the most prominent characteristics of ASD consists of difficulties in social communications, such as initiating or responding during a conversation as well as sustaining interactions with others appropriately in social situations (Koegel et al. 2008; Rao et al. 2008; White et al. 2007). Many children with ASD frequently experience social isolation from peers and tend not to engage in reciprocal interactions despite being in an inclusive environment (McConnell 2002; Rotheram-Fuller et al. 2010). The lack of appropriate social skills further inhibits the development of positive relationships with peers throughout the school years (Laushey and Heflin 2000). Findings of social skills groups also indicate that children with ASD do not automatically engage in social interactions with other children in a group setting unless social skills are specifically targeted (Banda et al. 2010; Wichnick et al. 2010). Social deficits inherent in children with ASD adversely affect their interactions with peers, rendering inclusion difficult and challenging to education practitioners. It is necessary to take advantage of inclusion and develop social skills interventions to facilitate appropriate social interactions for children with ASD. Such interventions may also begin early on in order to maximize social inclusion and acceptance in later years.

Peer-mediated intervention (PMI) is one evidence-based practice used to increase increasing social interactions between children with ASD and typically developing peers (Chan et al. 2009; Chang and Locke 2016; Ganz and Flores 2008; Wang et al. 2013; Watkins et al. 2015; Wong et al. 2015). In PMI, typically developing peers are trained to initiate and respond to social interactions to the children in question through various types of daily activities. Because PMI

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Published online: 16 February 2018

closely resembles daily activities in natural contexts, this type of intervention is particularly well suited for inclusive classrooms and has the potential to promote maintenance and generalization of acquired social skills (Hemmeter 2000; Trembath et al. 2009). Effective strategies used in PMI to improve social skills for children with ASD typically include the strategies derived from the principles of applied behavior analysis (ABA), such as modeling, prompting, reinforcing appropriate social behaviors (Oppenheim-Leaf et al. 2012; Trembath et al. 2009; Watkins et al. 2015), peer pivotal response training (Pierce and Schreibman 1995, 1997), and peer incidental teaching (McGee et al. 1992, 1999). Thus, it is essential to include ABA principles to guide interventions for individuals with ASD (Wong et al. 2015).

Previous PMI research indicates marked improvements in social skills for children with ASD in the areas of social initiations, communication, maintaining interactions, sharing, and turn-taking (Chan et al. 2009; Wang et al. 2013; Watkins et al. 2015). For example, Katz and Girolametto (2013) implemented a social skills training package to typical children and children with ASD as a group and increased the number and lengths of their interactions during play sessions. However, the play sessions involved unstructured play activities (e.g., playdough and blocks) that the children might not be interested in and required constant adult prompting to facilitate interactions during group training. Lee and Lee (2015) also used a PMI with effective teaching strategies to increase peer interactions for three children with ASD during snack time in a preschool. Results indicated that children with ASD maintained high levels of reciprocal interactions with their peers during snack time after the intervention. However, it is not clear whether the improved social interactions include initiations, responses to others, or both, as the researchers did not code these behaviors separately. Children with ASD often learn to respond well but continue to have difficulties initiating interactions, which is critical in engaging in meaningful relations with others (Haring and Lovinger 1989; Pierce and Schriebman 1995). Considering initiations and responses as distinctly different social skills, it is necessary to target and measure both skills as outcome variables to further examine the effects of PMI on social interactions for children with ASD.

Despite many advantages of using PMI in inclusive settings, PMI also has some potential difficulties and challenges in its implementation. The lack of treatment fidelity is the primary concern due to the fact that PMI relies on the ability of children to implement interventions as opposed to professionals (Horner et al. 2005). Other concerns include that PMI may potentially lead to stigmatization and social exclusion, and peers who implement the intervention may receive less instruction due to additional time required for peer training (Chan et al. 2009). In addition, most of the researchers using peer-mediated strategies may overlook emotional experiences or motivation of the target and peer participants (Koegel et al. 2012, 2013). Both the children in question and their typically developing peers are required to participate and behave in ways instructed by the experimenters, rather than engaging in the activities of their interest or preference. Taken together, PMI procedures need to be technically friendly for peer implementations. The selection of activities should be age and developmentally appropriate to all children involved. Instructional goals should be integrated in routine or play activities, and the activities themselves must be interesting and engaging to the children involved.

Previous research suggests that characteristics of activities and the number of social interactions for young children with ASD are correlated. For example, preschool children with ASD tend to have more social interactions when they are engaged in their preferred activities (Koegel et al. 1987). The activities should be predictable as well as more structured and organized so children know how they are expected to behave (Ferrara and Hill 1980; DeKlyen and Odom 1989). One type of activity that can serve as a medium for a PMI to enhance social interactions among preschool children in the inclusive classroom is LEGO® construction activities (Baron-Cohen et al. 2014).

The LEGO® play intervention is designed to maximize young children's learning motivation by embedding social interactions in LEGO® construction activities (LeGoff 2004). Many children, including children with ASD, are interested in highly structured, predictable, and systematic LEGO® play that results in natural reinforcement, such as the completion of a LEGO® model (LeGoff and Sherman 2006). In this social group, each of the three children is assigned a role with specific tasks and has to collaborate with one another to complete a collective construction project. The first person is designated as the "engineer," who describes the instruction and give directions, the second person as the "supplier," who finds the correct pieces, and the third person as the "builder," who puts these pieces together. Such a collaboration requires verbal and non-verbal communications, observations of others' work and the work of his own, and constant attention to the tasks, thereby creating opportunities for children with ASD and their non-ASD peers to engage in meaningful social interactions (Owens et al. 2008).

Studies also support that the LEGO®-based play activities have positive effects on social interactions, play, communication, and restrictive behaviors for children with ASD (Huskens et al. 2015; LeGoff 2004; LeGoff and Sherman 2006; Lindsay et al. 2017). LeGoff (2004) used a waitlist control group design involving seven groups of youth with ASD (N = 47; five to seven children per group, aged from 6 to 16) to evaluate a group Lego® play intervention. The intervention consisted of a division of labor among the participants, activity rules (instructed prior to sessions and

posted in the therapy room during sessions), and a therapeutic aide to facilitate interactions during sessions. Results indicated that children's social competence was improved significantly, including the number of social initiations, the duration of social interactions, and their repetitive and restricted behaviors. The improvements were maintained 24 weeks after the intervention. A subsequent study of the same intervention also reports a significant increase in children's social competence and the reduction of autism symptoms (LeGoff and Sherman 2006).

Owens et al. (2008) compared the effects of a group LEGO® play intervention and a social skills intervention as well as a non-intervention control group for children with high functioning autism and Asperger Syndrome. Children in the LEGO® group started off building quick and simple models in pairs or three with adult supervision. The social skills intervention involved the direct teaching of communication skills with designed social stories, group activities, and games. The results indicated that the LEGO® group improved significantly in social interactions, compared to the social skills group and the control group. The LEGO® group and the social skills groups also had significantly less maladaptive behaviors than the control group.

Previous research suggests that the LEGO® play intervention is potentially effective to improve various aspects of social behavior for children with ASD. However, some limitations exist that may limit the generalizability of the results. First, the LEGO® groups are comprised entirely of children with ASD. Without peers to model social interactions, constant adult monitoring and assistance is required to ensure appropriate and continuous interactions among the participants. The adult-directed interactive patterns may inevitably interrupt or disturb ongoing interactions among participants. Secondly, the intervention is conducted exclusively in clinical settings by clinical professionals. It is not clear whether the intervention can be carried out by non-clinical professionals and whether the same improvements of social behaviors can be observed in the natural environment. Therefore, it is necessary to include typical peers in the LEGO® group supervised by non-clinical personnel in a less restrictive environment in order to promote generalization of acquired social skills. Thirdly, the reviewed PMI and LEGO® intervention studies do not measure the number of prompts provided by peers throughout the intervention sessions. Measuring the number of peer prompts provided to children with ASD may serve as an indicator for skill acquisition and independence in group play. Finally, the LEGO® studies only evaluate the intervention for school-aged children with ASD. No younger children prior to school age are included in the studies.

Current available research suggests PMI as a potential intervention option to improve social functioning for children with ASD, and the LEGO® play intervention also indicates that LEGO® play activities may serve as a medium for such a purpose. Previously, no formal attempt had been made to conduct an empirical investigation combining PMI with LEGO® play as an intervention to improve children's social interactions in inclusive settings. The purpose of this study was to investigate the effects of this approach on social interactions for preschool children with ASD in an inclusive setting. The following research questions were addressed: (a) did the number of social interactions, including social initiations and social responses, change in children with ASD as a result of the peer-mediated LEGO® play intervention? (b) did the number of peer prompts provided during intervention change over time? and (c) did the teachers perceive that the intervention was feasible, acceptable, and effective?

Method

Participants

Children with ASD

Three male preschool children (Yangyang, Xinxin, Youyou), aged 4–6 years old, participated in this study. They were enrolled in a privately funded inclusive preschool in Beijing, China. They were diagnosed as ASD by physicians or clinical psychologists with the Chinese version of the *Childhood Autism Rating Scale-2nd edition* (CARS-2; Lu et al. 2004; Schopler et al. 2010), and the Diagnostic and Statistical Manual of Mental Disorders (DSM-5; American Psychiatric Association 2013). The scores of CARS-2 categorize autism into severe ($\geq$ 37), mild to moderate (30–36.5), and minimal to no symptoms (15–29.5).

Yangyang was 6 years old at the time of the study. His CARS-2 score was 30, and his IQ score was 112 assessed from the Chinese version of the *Wechsler Preschool and Primary Scale of Intelligence-Fourth Edition* (WPPSI-IV; Li et al. 2011; Wechsler 2012). He recognized approximately 100 Chinese words and could read story books with pictures and words for his age. Observations of his play indicated that he preferred solitary play and seldom engaged in parallel play or in group activities even though he was seated with other peers. He spent most of his time playing with toys by lining them up repetitively. He had very few interactions with his peers during observations of free play. According to his teacher, Yangyang often had tantrums due to unexpected changes in daily routines, transitions from activities in different rooms, and any denial of his requests.

Xinxin was 6 years old with a score of 32 in CARS-2 and an IQ of 104 measured by WPPSI-IV. He did not have any interactions with peers in the classroom, and other children tended to avoid or ignore him during free play. He used no more than three-word phrases to make requests from adults

only. However, his reading level was equivalent to children at his age and he also recognized approximately 100 Chinese words. He was observed to read books during free play time. He responded to one-step directions given by adults but usually needed prompts to complete the tasks. His teacher reported that he frequently cried, threw things, and avoided challenging tasks when presented to him. He also had tantrums for changes of daily schedules or any re-arrangement in the environment.

Youyou was 4 years old with a score of 30 in CARS-2 and an IQ of 107 in WPPSI-IV. Observations of his play indicated that he showed interests in peers and attempted to initiate interactions with them, but his social initiations usually involved giving directions to others. When other children initiated interactions or responded to him, he did not respond but ignored them. Most of the children were not willing to play with him but some children occasionally followed his lead. Youyou liked to read books and play with building blocks, construction trucks, and cars during free play time. He followed verbal directions given by adults and used phrases to make requests. He could recognize at least 50 Chinese words.

Typically Developing Peers

We asked the head teachers to nominate several typically developing peers and the children with ASD in their classrooms to participate in this study. To meet criteria as a peer, a child must (a) have good verbal communication skills, (b) have good social skills, (c) regularly attend the school, (d) be in the same classroom as the child with ASD, (e) be willing to serve as a peer for this study, and (f) show an interest in interacting with the child with ASD. We included three girls and 10 boys, aged 4–6, as peers in this study.

Settings and Materials

This study took place in a specified LEGO® area in the classrooms of the preschool. The classroom ($10 \times 12 \times 4$ m) contained multiple play and work areas, including a reading area, a dollhouse area, a LEGO® area, a construction truck area, and a drawing area. A typical day (9 h) in the preschool included two 45 min group instruction sessions (i.e., reading, math, science, and art curricular activities), three 60 min free play sessions, two 45 min outdoor recesses, a 40 min lunch, a 90 min nap, and two 30 min snack breaks. Each classroom had one head teacher, three teacher's aids, and 20–25 children, including three to five children with ASD or mild intellectual disabilities. There were seven classrooms in this preschool, including two 3 year-old toddler classrooms, two 4 year-old preschool classrooms, two kindergarten classrooms, and a slef-contained classroom.

The LEGO® area had a space of 3×4 m displayed with the DUPLO® blocks (larger-sized blocks designed for children from 2 to 6 years). All target children with ASD had experience of playing with the DUPLO® blocks during free play time prior to the study. Based on observations of their selections of the LEGO® models, we used the most frequently chosen themes, including "Cars," "Houses," "Robots," "Animals," and "Amusement." Each theme had specific model builds. Our observations indicated that the target children with ASD usually stacked some pieces or line them up, but rarely made other patterns, or completed the entire models during their free play. We photographed the completed tasks of the models and printed them out in A4 size paper as visual cues used in the intervention sessions.

Experimental Design

We employed an A-B-A-B reversal design as the primary design combined with a non-concurrent multiple probe design across three participants as the secondary design (Kennedy 2005) to examine the effects of peer-mediated LEGO® intervention on children's social interactions during free play. Following a stable baseline performance, the intervention was introduced to the first child. As the first child's social interactions displayed an ascending trend over five intervention sessions, we implemented the intervention to the second child, and the same criterion was used to determine when to introduce the intervention to the third child. The intervention was withdrawn with a return to baseline once the child's social interactions reached a stable performance across three consecutive sessions. After the second baseline, the intervention was re-introduced to the child until the stable performance. The other two children also followed the same sequence.

Procedure

Baseline

Baseline data on social interactions were collected during 30 min unstructured free play time in the morning. In the beginning of each free play session, the head teacher announced to all children that it was free play time and they could play with any materials available in any area of the classroom. All play materials were placed on the shelves or drawers at children's eye level and easily accessible to them. Each participant with ASD and his classmates were engaged in the play activities of their choice. All three target children were observed to enter and stay in the block area and played with blocks during baseline. However, they only stacked or lined up blocks and did not build any other patterns.

The teachers monitored children's activities and responded to their requests during free play. If any child

did not follow classroom rules or had conflicts with other children, a teacher would come up to resolve the problems. A free play session ended with a bell ring and the teachers would announce the next transition to the children.

Pre-intervention Peer Training

We conducted the pre-intervention training sessions for the 13 participating peers one month before the intervention took place. Two children were paired to receive one 25–30 min training session. The training included basic rules in LEGO® play, opportunities to respond and initiate to an interaction, and how to provide visual or physical prompts, modeling, and reinforcement to the target children with ASD (See Table 1 for training objectives, examples, and mastery criteria). We utilized role play (the experimenter as the target child with ASD) to simulate the intervention with in-vivo modeling and coaching in peer training. Each peer was assessed with a checklist with pre-determined mastery criteria. The training and role play activities continued until both peers achieved all mastery criteria. A review training session took place one day prior to each intervention session. We pulled out two peers, who were available and willing to participate in the intervention session the next day, from one of the free play sessions to conduct the review training. The training protocol was the same as the first training session.

Intervention

The intervention sessions were conducted in the specified LEGO® area in the classroom. We displayed textual prompts on the wall as reminders for all children. Some examples of the textual prompts were as follow. "If you have any questions, ask your peers." "One Engineer at a time and follow the Engineer's directions." "If you get help from others, remember to say 'thank you.'" "When talking to a person, look at his eyes."

Each intervention session lasted for approximately 40 min and was conducted with the following steps. Step 1 (5 min): Showing the available LEGO® task models. The experimenter showed all three children (one target child with ASD and two peers) all LEGO® task models for that session. Each child chose one model from the pool. Step 2 (2 min): A brief review for the peers. The experimenter reviewed the modeling and prompting techniques to assist the target child with their peers. Meanwhile, the experimenter asked the target child to observe the visual cues of his chosen model. Step 3 (3 min): Explaining roles. The experimenter explained the three roles (i.e., engineer, supplier, and builder) to the target child, while other two peers observed the visual cues of the task models of their choice. Step 4 (10 min): Deciding roles and building the first model. When the first model was selected, the experimenter discussed and decided roles. Each child was assigned one of the three roles for each task model in the order of the engineer, the supplier, and the builder. When a task model was complete, the children were excited and celebrated their accomplishment naturally by clapping hands and giving high fives to each other (e.g., "Very good! Yay! Yay! Yay!"). Such a celebration was a typical way the children celebrated their achievements in group activities in the classroom. Step 5 (20 min): Switching roles for other two models with adult assistance. After the brief celebration ceremony for the completion of the first task, the supplier then switched to the engineer role for the second model. Each child must rotate to a different role for each task model. A brief celebration ceremony was held as a natural reinforcement each time a model was complete.

During block building (Step 4 and 5), the experimenter sat behind the children supervising their activities. If a child was off task for more than 15 s, the experimenter reminded him to focus on the ongoing task. When the target child did not respond appropriately, and the peers did not provide any necessary prompts within 10 s, the experimenter provided a verbal reminder to the peers. For example, if the target child served as the builder but did not respond to a request from the engineer, the experimenter reminded both peers serving as the engineer and the supplier, "You can point to the item you requested or show the Builder what to do." If any conflict occurred, the experimenter asked all three children to read the textual prompts on the wall and led a brief discussion, if necessary. If a child was non-compliant or displayed problem behavior for more than 5 min, an intervention session was terminated. No session was terminated under such a condition in this study. Each intervention session ended following the celebration ceremony for the completion of the third task model.

Response Definitions and Data Collection

The primary dependent variables consisted of social initiations and social responses by the target children with ASD. An instance of a social initiation was defined as any verbal or non-verbal social behavior displayed by the target child with ASD to a peer or to the group as a whole without any preceding interaction. Examples of verbal social initiations included saying, "I need a yellow square block", "It's my turn," or "This model is easy to build." Examples of non-verbal social initiations included tapping a peer's arm to get attention, pointing to a DUPLO® block, or handing a DUPLO® block to a peer. An instance of a social response was defined as any verbal or nonverbal social behavior by the target child that acknowledged, related to, or complied with a preceding initiation (Lee and Lee 2015). Some examples included answering a peer's question or responding to a peer's requests.

Table 1 Content and mastery criteria of pre-intervention peer training

Content	Objectives	Instruction	Examples	Mastery criteria
LEGO® play	Naming LEGO pieces	Labeling color, shape, function, features of LEGO® pieces	"Give me a red square rectangle LEGO® brick"	Less than 3 mistakes in play
	Building skills	Assembling the pieces in a right way; building a LEGO® model based on the visual cues	Teach peers to build from the bottom to the top	Less than 3 mistakes in play
	Keeping roles	Explaining the job of three roles ("Engineer", "Supplier", "Builder") in collaboration and responses to overstep performing	If the "engineer" was trying to find materials or build, the others would remind him	Less than 3 times of breaking rules
Social Skills	Response	Joint attention, answering questions, providing help, responding to comments and requests	Take action as soon as getting the directions	Less than 3 times of inappropriate responses
	Initiation	calling peer's name looking at peers, asking questions, making requests or comments, initiating joint attention, seeking help	Ask peers for help, "could you help me hold this?"	Less than 3 times of inappropriate initiation
Intervention Strategies	Modeling	Demonstration of a desired action or verbal expressions	Peer says "you can say 'Please give me a blue square'"	Used at least 1/2 times to the applicable opportunities
	Prompt	A hint or reminder for a desired action or verbal expressions, including verbal, pointing, gesture, and physical prompt	Peers touch another participant's arms to look at the visual cue	Used at least 1/2 times to the applicable opportunities
	Reinforcement	Praise, patting, hugging, clapping hands when he/she completes a desired action	"You are doing great! Thanks for being so helpful!"	Used at least 1/2 times to the applicable opportunities

Social initiations and responses were coded if the verbal or nonverbal communications were clearly directed to a peer through eye contact, eye gaze, body orientation, or any attempt to seek attention (e.g., name-calling, holding an object up). A series of communication or requests expressed by a child without pauses within 5 s were counted as one instance. For example, the child said, "Can you bring me a blue square? Oh, help me. It is broken!" This was coded as one instance of social initiation. Prompted initiations and responses were not counted, neither were repetitive vocalizations or echolalia (e.g., repetitively saying "come on"). Behaviors directed to adults, self-narration (e.g., talking to objects), and communications not clearly directed to a peer were also not coded.

The secondary dependent variable was the number of prompts delivered by the peers during the intervention. A prompt was defined as a verbal instruction (e.g., "Xinxin, Engineer is talking to you"), a gestural prompt (e.g., pointing to a specific block), and an act of physical guidance (e.g., holding the child's hand to build a block) provided by a peer to the child with ASD during the intervention sessions. If more than one type of prompts were used at the same time for a particular response, it was recorded as one prompt. The peers used either verbal instructions, gestural prompts, or both, and rarely used physical guidance in this study.

All baseline and intervention sessions were videotaped for the purpose of data collection. A video camera was set up in the corner of the classroom facing the LEGO® area for each session. Data were recorded via the paper-and-pencil form.

Social Validity Measures

After the intervention was completed, the experimenter conducted a survey with the three teachers who worked with the target children. The teachers completed an eight-item questionnaire developed by the researchers to assess the feasibility, acceptability, effectiveness, and satisfaction of the intervention. Each item was rated on a 5-point Likert scale (1 = not agree to 5 = strongly agree). Item 1 was related to the participants' interest in the activities. Item 2–4 pertained to the effectiveness of the intervention for the target children and peers (i.e., activity engagement, social initiation with peers and social response to peers). Item 5 regarded the acceptability of the intervention. Item 6 to 7 measured the feasibility of the intervention (i.e., time, location and procedure). The last item assessed the overall perceived helpfulness for the target skills in the intervention. Each item also had an open-ended question for responders to provide more detailed information on the question. In addition, all 16 participating children were interviewed after the intervention. The interview included four questions asking their satisfaction with the intervention, their attitudes toward peers, and their willingness to play together in other activities and settings.

Procedural Integrity and Interobserver Agreement

To ensure that the experimental procedures were implemented accurately, procedural integrity was evaluated weekly by two supervisors. A total of 36% of the experimental sessions were assessed with procedural integrity. These sessions were equally divided across all conditions and all children with ASD. The supervisors checked the procedural

Table 2 Procedural integrity checklist

Items	Yes/no
Baseline	
1. Did the teacher tell the children it was free play time in the beginning of each session?	
2. Were the play areas arranged in a way that all children could move freely and safely?	
3. Were the play materials in each play area placed at children's eye level and easily accessible to children?	
4. Did the teacher tell the children to clean up all items 5 min before the end of each session?	
5. Did the teachers monitor if children interacted with each other nicely?	
6. Did the teachers resolve children's conflict, if any?	
7. Did each session last for 30 min?	
Intervention	
1. Did the time allocation follow the schedule?	
2. Did the intervention session start with showing them task models?	
3. Did the experimenter provide enough models to choose from?	
4. Did the experimenter review the social skills and PMI strategies before starting activity?	
5. Did the experimenter explain the three roles to the targeted child clearly?	
6. Did the experimenter guide the children to choose the tasks, building sequences, roles to play by themselves?	
7. If there was a conflict, did they settle with appropriate guidance?	
8. Did the children complete the tasks collaboratively based on the division of labor?	
9. Did the children have a celebration ceremony each time a model was complete?	
10. Did they have 30 min to build the tasks?	
11. Did the peers initiate interactions appropriately?	
12. Did the peers respond to interactions appropriately?	
13. Did peers provide prompts and reinforcement appropriately?	
14. Did the experimenter provide verbal prompts to the peers if the peers did not prompt the target child within 10 s?	
15. Did the researcher provide appropriate support and supervision during intervention?	

integrity in baseline and intervention with a brief checklist containing all planned procedural components (See Table 2 for the checklist). If the intervention steps were implemented as described in the checklist, a "Yes" was recorded for that step; if not, a "No" was recorded. Procedural integrity was calculated by dividing the number of accurate implementations (marked with "Yes") and the total number of implementations in the evaluated areas and multiplying 100. The percentage of procedural integrity was 100% in all sessions observed.

To assess interobserver agreement (IOA), the experimenter and two other graduate students who were naïve to the purpose of the study independently recorded the target children's social initiations and responses as well as peer prompts from video recordings for 33% of baseline sessions and 40% of intervention sessions. IOA was calculated by dividing the number of agreements on the occurrences of social initiations, social responses or peer prompts by the total number of agreements and disagreements and multiplying by 100. The average of IOA for the frequency of social initiations was 88% (range 82–100%) and 91% (range 88–100%) for the frequency of social responses. IOA for the number of peer prompts averaged 95% with a range from 92 to 97%.

Results

Figure 1 depicts the frequency of social initiations and responses as well as the number of prompts across conditions for three children. Overall data showed that social initiations and responses were at a low level during baseline but increased to a relatively high level with an ascending trend under the intervention conditions for all three children. The peers delivered a steady level of prompts with a gradual descending trend throughout intervention sessions for each child. The data clearly indicated a functional relationship between the peer-mediated LEGO® intervention and the social interactions of children with ASD.

Social Initiations and Responses

Yangyang had a very low level of social initiations (range 0–1) and responses (range 0–0) during initial baseline. As the intervention was introduced, social initiations and responses showed immediate, slight increases and to a relatively high level with ascending trends (social initiations: range 4–28; social responses range 14–40). Both social initiations and responses immediately returned to baseline performance (social initiations: range 0–3; social responses: range 0–5) during the second baseline. The data showed an immediate increase and ascended to a relatively high level when the intervention was re-introduced (social initiation range 17–38; social responses range 10–41). There was no data overlap between baseline and intervention conditions.

Xinxin had no social initiations and responses in the initial baseline. His social responses gradually increased from a low level to a relatively high level (range 4–28), while his social initiations increased to a slightly high level and maintained at the same level (range 2–7) under the first intervention condition, compared to the initial baseline. When the intervention was removed, his social interactions returned to the baseline level with a range from zero to one social initiation and a range from one to four social responses per session. The frequency of social initiations increased to a slightly high level (range 2–11) while that of social responses increased abruptly from the second session and maintained at a high level (range 4–34) under the second intervention condition. No overlap existed between the first baseline and the first intervention. However, there was a 16.67% overlap between the second baseline and the second intervention.

Youyou also had a low level of social interactions during the initial baseline (social initiations range 1–6; social responses range 0–3) and the second baseline (social initiations range 4–9; social responses range 4–10). Under the first intervention condition, both social initiations and social responses showed immediate increases with ascending trends (social initiations range 13–39; social responses range 22–41). The data also showed immediate increases in the beginning session of the second intervention condition and maintained at a high level for social initiations (range 27–34) and for social responses (range 26–37). The data reflected immediacy of the intervention effect and had no overlaps between the baseline and intervention conditions.

Peer Prompts

During the first baseline, no prompts were provided to the target children. Yangyang was prompted 13–22 times per session in the first intervention and then 12–16 per session in the second intervention condition. Additionally, Yangyang received a total of 4 prompts (range 0–2 per session) from peers during the second baseline.

Xinxin required 22–33 prompts per session during the first intervention, a 0–1 prompt per session (total 2 prompts) during the second baseline, and 21–29 prompts per session during the second intervention.

Youyou was assisted with 11–18 prompts per session during the first interventionand, 10–16 prompts per session during the second intervention. Youyou received a

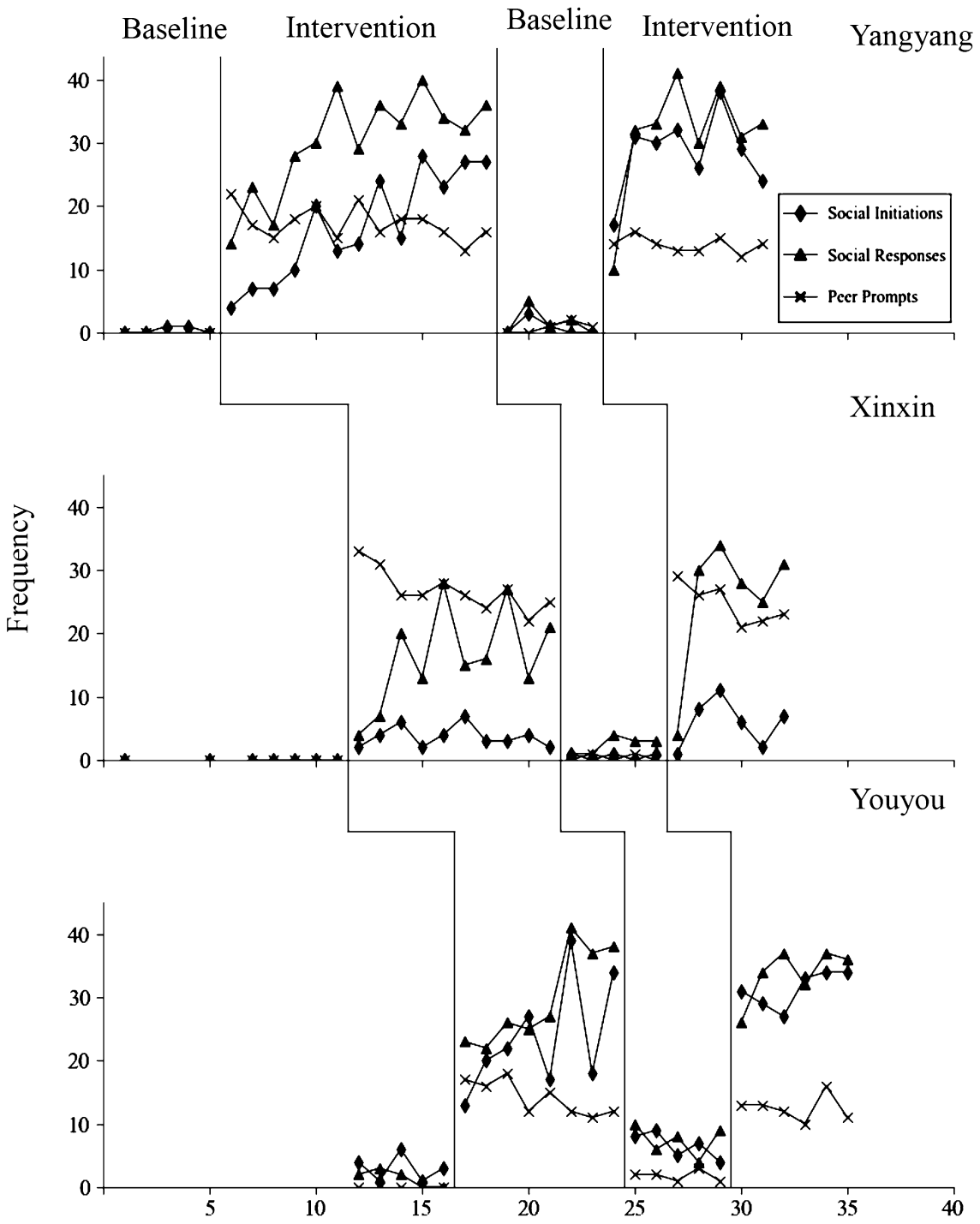


Fig. 1 Frequency of social initiations, social responses, and peer prompts during 40 min sessions across conditions and participants

total of nine prompts (1–3 per session) from peers during the second baseline.

Overall, peer prompts provided for each child were maintained at a stable level with a gradual descending trend throughout the intervention sessions. Xinxin required relatively more frequent prompts than Yangyang and Youyou, but the number of prompts also displayed a gradual descending trend from the first intervention to the second intervention.

Social Validity

Results of the teacher survey conducted by the first author showed that the average ratings were 5 ($SD = 0$) on participants' interest to the intervention, 4.33 ($SD = 0.52$) on intervention effectiveness, 5 ($SD = 0$) on acceptability, 4.56 ($SD = .53$) on feasibility, 5 ($SD = 0$) on overall satisfaction and of the intervention to improve target children's social interactions. All three teachers reported positive changes in the target child's social skills during and after the intervention in their classrooms. They also noted that the most significant change was that the target children showed more interest in peers and interacted with them more frequently. The second author conducted the children's interviews. All children, including children with ASD and the participating peers, reported a high level of satisfaction and enjoyment of the LEGO® activities as well as a positive attitude change toward peers. They also reported their enjoyment of the friendship and strong willingness to play together in other activities.

Discussion

The study was the first attempt to evaluate the effects of a peer-mediated LEGO® play intervention on social initiations and social responses for three preschool children with ASD during group play in an inclusive setting. The data showed a strong functional relationship between the intervention and the children's social initiations and responses in group play. Additionally, a high level of procedural integrity in peer mediation strengthened the experimental control of this study.

Social Initiations and Responses

The results of the study were consistent with previous studies that PMIs improved social interactions for children with ASD in inclusive settings (Katz and Girolametto 2013; Lee and Lee 2015). Data on social interactions suggested that the intervention was effective in increasing both social initiations and responses for Yangyang and Youyou. For Xinxin, the intervention improved his social responses but social initiations remained a challenge in his repertoire. Observations suggested that Xinxin was sometimes distracted and required adult reminders to maintain on task during intervention sessions. Adequate attending skills might be necessary to obtain maximum intervention effects. Additionally, data also indicated all children had a higher level of social responses than social initiations, indicating social initiations are more difficult to teach to children with ASD. Our observations suggested that the role of the engineer provided more opportunities for children to engage in social initiations than other roles in the intervention. Therefore, it is possible to assign a leadership role to a child with ASD that may create opportunities for social initiations with other children.

The frequency of social initiations and social responses returned to near baseline levels for all three children with ASD during the second baseline, suggesting the structured peer LEGO® activities with adult monitoring were necessary to occasion social interactions in a play setting. Simply arranging children with ASD to stay with their peers in unstructured free play without instruction was not sufficient for social interactions to occur. Therefore, in order for a PMI to be effective, it may be necessary to provide more structured activities with adult support to facilitate peer mediation in a play setting, especially when young children serve as peers to implement the procedures.

The results of the study provided evidence to support the effectiveness of the peer-mediated LEGO® intervention on improving social interactions for preschool children with ASD in an inclusive setting. Similar to previous studies, we also used structured LEGO® construction activities in a small group setting (LeGoff 2004; LeGoff and Sherman 2006). The use of the DUPLO® sets was age and developmentally appropriate for the children in our study. Different from previous LEGO® studies conducted by therapists in clinical settings, we used peer prompts and visual supports in the LEGO® building activities, and thus, the adult assistance was minimized. With PMI, the peers provided modeling of appropriate social behaviors and delivered prompts to the target child engaging in group play activities. The visual supports provided pictures of completed models to facilitate independence for young children to engage in the construction activities. It is possible that the strategies based on the ABA principles used in peer mediation are a key to the success of the intervention, as previous studies using only LEGO® therapy without peer mediation indicated inconclusive or mixed effects on children's social behavior change (Owens et al. 2008). However, this assertion warrants further investigations.

In contrast to previous studies where the therapist played a more directive role in the LEGO® play, the adult in our study played an assistant role by guiding the children to make choice of preferred models and transition to a different role with another task model. Additionally, we conducted the intervention in a natural setting, a regular preschool with the presence of other children in the classroom, but not in a relatively restrictive clinical setting.

Generalization is an important element in any intervention and needs to be programmed in the procedure (Stoke and Baer 1977). We incorporated generalization in the intervention procedure in the following ways. Each child rotated into all three roles (i.e., the engineer, the supplier, and the builder) with specific task responsibilities for each intervention session. The target children had to follow three sets of

different responsibilities in collaboration with others. We also included a total of 13 peers rotated across the intervention sessions, so the target children worked with different peers across sessions. We also had a total of 40 LEGO® DUPLO® models available for the study and systematically alternated the models to provide only ten models for the children to select in each session. Thus, the children did not repeatedly select the same models.

Besides improving social interactions with others, it is important to expand their community of interest in leisure and play activities as a long-term goal for children with ASD (Greer and Ross 2008). This is relevant to the core deficit of restrictive interests and activities inherent in ASD. As discussed, many children with ASD may prefer LEGO®-related activities as these activities are structured with predictable outcomes. The intervention implemented here not only introduced a diverse variety of LEGO® models to increase the diversity of play behavior but also expanded LEGO® play as a cooperative and fun activity in a group with peers sharing the same interest.

Peer Prompts

Peer prompt data showed that that the peers provided a stable level of prompts with a gradual descending trend for each child throughout the intervention sessions. For Yangyang and Youyou, the number of prompts remained at a relatively low level while their social interactions were ascending to a relatively high level, indicating social interactions occurred naturally via structured LEGO® activities with minimal prompts. It is possible that the structured LEGO® activities, the pictures of completed products, and the brief celebration provided motivation with sufficient opportunities to maintain and strengthen continuous engagement and social interactions in the group. The gradually decreased amount of peer prompts during intervention sessions also suggested that the children's play skills were improved over time through the intervention.

Compared to the other two children, Xinxin required more frequent prompts. However, the gradual descending trend of peer prompts across intervention sessions suggested that his play skills were also improved throughout the intervention. The potential reasons for Xinxin to require more prompts may be due to the fact that his attending skills were relatively weak. Therefore, he needed more assistance to remain on task, attend to visual supports, and complete assigned tasks accurately. It may help to maximize intervention effects by first training his attending skills prior to the intervention.

Interestingly, we observed that some peers trained for the intervention began to engage in social interactions and provide prompts to children with ASD during the second baseline in a free play setting. Although the number of peer prompts was low, this suggested a possibility of training peers to engage in social interactions with children with ASD in unstructured free play settings outside of intervention for future studies.

Social Validity

Results of social validity indicated that children with ASD and their typically developing peers were highly interested in the LEGO® activities used during intervention sessions. The results of the teacher survey also supported the acceptability, feasibility, and satisfaction, suggesting the potential of replicating the same procedure in other inclusive preschools in a larger scale.

Strengths

The strengths of the intervention in this study were as follows. First, the procedure improved both social initiations and social responses, which were two different behaviors in the social repertoire for children with ASD (Pierce and Shriebman 1995; Haring and Lovinger 1989). This probably resulted from the structured division of labor requiring skills in social initiations (e.g., the engineer who gave directions) and social responses (e.g., the supplier and the builder followed directions to provided pieces and put pieces in place, respectively). Secondly, the procedure set the occasion for generalized collaborative play behavior. As described above, the children were exposed to different role responsibilities, peers, and task models for the purpose of generalization. Thirdly, the procedure emphasized children's interest, as advocated by Koegel and Koegel (1995). The LEGO® materials were preferred activities for all children who volunteered to participate in this study. Each child was provided opportunities to choose one of their preferred task models for each session. Fourthly, adult monitoring was required but the assistance was minimized via the utilization of the PMI strategies. The adult played an assistant but not a directive role in the intervention. Fifthly, the intervention used peers to provide modeling and prompts, which approximates natural social interactions among children. Sixthly, the intervention utilized natural reinforcement as consequences for the target social behaviors and for the completion of each task model. Their social behaviors were reinforced naturally by peers' responses (e.g., a request for a piece of block resulted in the delivery of the piece requested; delivering a requested piece to a peer was reinforced by the peer saying "thank you"). Sixly, the intervention was conducted in a school environment with the presence of other children and teachers in the classroom. The target children were blended with peers in a natural play setting, and therefore, the potential problem of social isolation or stigmatization was minimized. Lastly, the training for the peers used one of their free play session

and did not take them away from their regular instructional time or interrupted their daily schedule.

Limitations and Future Research

Limitations of this study consisted of the lack of follow-up maintenance sessions and the measure of problem behaviors across conditions for children who displayed them. It is necessary for future research to include short-term and long-term follow-up sessions to evaluate the maintenance effect of the intervention. It is also necessary to include any problem behaviors (e.g., tantrums) or inappropriate play behaviors (e.g., stereotypic behaviors) as dependent variables in future studies.

Another limitation was that the experimental design did not allow component analysis. Therefore, it remains unclear whether both the PMI and LEGO® were necessary or which component in isolation produced stronger effects on the target social behaviors. Future researchers can consider conducting component analysis to examine the effects of LEGO® intervention on children's social interactions with and without the component of PMI.

The experimental design of this study also met the quality indicators for reversal designs recommended in Kratochwill et al. (2010). A relatively minor limitation was the delay of baseline data collection for the third child in the multiple probe across participants design as a secondary design of this study.

In addition, the teachers may be trained to implement the intervention after the study was completed in order to sustain the intervention effects in a long run. Future researchers can consider providing teacher training and evaluate the effects of the same intervention on children's social behaviors with teacher iplementations.

Implications

The results of this study have important implications for inclusion of preschool children with ASD. Many children with ASD, especially those with high-functioning ASD, are enrolled in regular preschools. The peer-mediated LEGO® play intervention used in this study provides an option for teachers to facilitate peer social interactions with structured activities in an inclusive environment. Using peers as a helpful resource, this intervention also serves as a medium for children with ASD to build up positive peer relationships from early on. The implementation of this intervention is also relatively easy and fits into the daily schedules of regular preschool settings.

Acknowledgments This research was supported by 2016 Comprehensive Discipline Construction Fund, Faculty of Education, Beijing Normal University.

Author Contributions XH conceived of the study, participated in the design and coordination and drafted part of the manuscript. QZ conceived of the study, participated in the design and coordination and analyzed the data. GTL analyzed and interpreted the data, drafted part of the manuscript, and revised the entire manuscript critically. All authors read and approved the final manuscript.

Compliance with Ethical Standards

Conflict of interest All authors involved in this study do not have any interests that might be interpreted as influencing the research.

Ethical Approval All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Informed Consent Informed consent was obtained from all participants (teachers, children and their parents) included in the study.

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